

HEC Lausanne

Financial Econometrics

Can Oil Market Volatility Predict U.S. Financial
Market Volatility through instantaneous comovements
?



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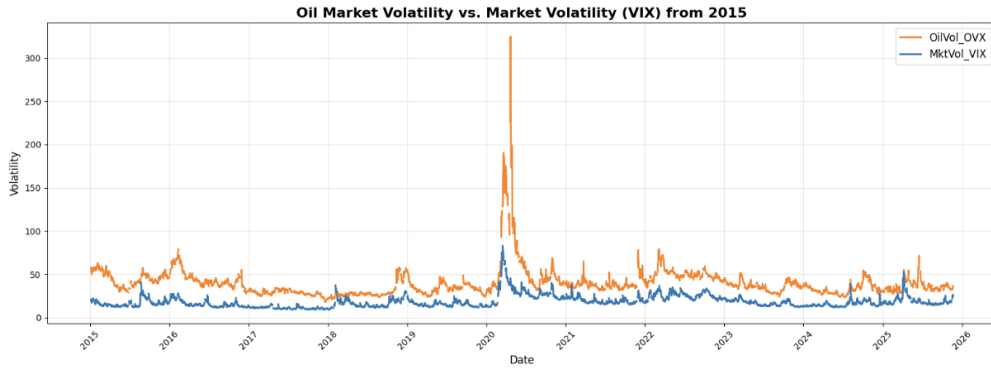
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Abstract

This study investigates whether oil market volatility can predict U.S. financial market volatility through instantaneous comovement. Using daily data on key financial and macroeconomic variables, we model the logarithm of the CBOE Volatility Index (VIX) as a function of the oil market volatility index (OVX) and several control variables capturing financial stress, uncertainty, and macroeconomic conditions. The results of the specified OLS regression indicate a strong and statistically significant simultaneous relationship between oil and financial market volatility. These findings support our hypothesis that oil market uncertainty contributes to short-term fluctuations in systemic financial risk. They highlight the interdependence between commodities, specifically oil in our study, and financial markets and suggest that oil volatility can serve as a useful early indicator for monitoring financial market instability. We will also see within the discussion of extension part, the uncontrolled problems highlighted by our study, and the technical limitations that arise for our results.

Figure 1: Market volatility and oil market volatility from 2015



1 Introduction

Oil has occupied a central place in the world economy for many decades. As one of the most strategic assets, it supplies growth, influences production costs and is conditioning, in a large scale, geopolitical stability and relations worldwide. Beyond its energetic dimension, the price of oil constitutes a synthetic index of both economical and geopolitical tensions. Therefore, fluctuations in oil prices should be reflected well further than the only commodities market, affecting for example the forecasts of economic agents, the value of financial assets and finally, the global volatility of markets. We will thus carry out in this context a study focused on direct comovements and not on lagged comovements for several reasons. First, we proceed here under the assumptions of liquid and thus efficient markets. The large capitalizations and high trading volumes lead us to believe that instantaneous comovements would allow us to avoid the loss of information on the causality of the relationship that a lagged comovement would entail. Moreover, this study aims to identify the instantaneous relationship between these two metrics in order to use this relationship for immediate risk management; therefore, lagged estimations would not be relevant. For the choice of variables, the U.S. market's VIX (the so-called "fear index") captures these risk expectations. The VIX therefore provides an indication of market volatility based on implied volatility. We will thus use this variable as a proxy for market volatility. The oil market,

by its central macroeconomic role, but also by its exposure to supply, demand shock and other tensions, constitutes a potentially huge channel of systemic risk transmission. We will therefore use, for the volatility of the price of the oil market, the VIX calculated by the FRED on the oil market. Regarding the econometric methodology, we will first carry out a simple regression to which we will add relevant control variables. We will then look at the optimal functional specification based on the specification tests seen in class, and finally move on to more complex estimation methods, such as finding an instrumental variable to control for other external issues (simultaneity).

In this context, we raise the following question: Can volatility in the oil market immediately forecast volatility in U.S. financial markets? To address this question, we propose the following hypothesis:

- H_0 : Volatility in the oil market does not increase systemic risk in the financial market.
- H_1 : Volatility in the oil market increases systemic risk in the financial market.

2 Data Selection

2.1 Selection of Control Variables

The dataset used in this project is sourced from the Federal Reserve Economic Data (FRED) database of the St. Louis Federal Reserve. This source was selected for its institutional reliability, its extensive macroeconomic and financial coverage, and its convenient free access through a dedicated Python API. Our initial exploration spanned 33 variables, which we subsequently organized into thematic sub-groups. The analysis focuses exclusively on the U.S. economy, given its global economic and financial influence. However, not all variables were retained in the final specification due to redundancy or limited relevance.

2.1.1 Economic Uncertainty

This group captures aggregate financial stress and uncertainty in U.S. markets. It includes composite indices such as the FSI, NFCI and ANFCI, which summarize liquidity conditions, volatility, credit spreads and market-wide risk factors.

2.1.2 Credit Risk and Market Doubt

This category reflects borrowing conditions, default premia and investor risk perception. It includes measures such as corporate bond spreads, high-yield risk premia and interbank tension indicators like the TED spread.

2.1.3 Interest Rates and Monetary Policy

This group represents the stance of monetary policy and expectations about future Federal Reserve actions. Yield-curve spreads are included here, as they serve as forward-looking predictors of macroeconomic slowdowns or recession risks.

2.1.4 Economic Health and Inflation

This category provides the broader macroeconomic environment. It includes inflation metrics, labor-market indicators, industrial production measures and recession-regime variables.

2.1.5 Specific Influence Variables

This final group captures external and policy-driven forces. It includes exchange rates, the Federal Reserve balance sheet, economic policy uncertainty indices and global commodity prices such as copper.

2.2 Dataset Cleaning

Some variables in the dataset are reported at lower frequencies (weekly or monthly), which naturally leads to missing values on days without official announcements. For example, monthly indicators such as CPI exhibit a single value per month, with all other daily observations initially recorded as missing.

To preserve the daily granularity required for the analysis, we filled missing observations for these low-frequency variables using the most recently published value. In practice, for a monthly series like CPI, all daily entries of a given month were assigned the value released at the beginning of the month. After applying this forward-filling procedure, the remaining missing values—primarily due to exceptional shocks or isolated reporting anomalies—were removed from the dataset.

3 Baseline Model and Econometric Issues

3.1 Baseline OLS specification

We start with a simple linear relationship:

$$\text{MktVol_VIX}_i = \beta_0 + \beta_1 \text{OilVol_OVX}_i + \varepsilon_i. \quad (1)$$

We expect β_1 to be positive, as an increase in oil market volatility is likely to translate into higher financial market turbulence.

The model is initially estimated using ordinary least squares (OLS) with non-robust standard errors.

As we can see in the following table, the correlation between market volatility and oil volatility is statistically significant and confirms that studying more about a potential causal relationship makes sense. Despite the high coefficient which seems to indicate that each % of oil volatility leads to an increase of 0.2684% of market volatility, we don't expect that coefficient to represent a causal effect because of several econometric issues. The corresponding OLS regression is summarized in Table 1.

Table 1: Baseline OLS: Market volatility on oil volatility

	Coefficient	Std. error	t-stat
Constant	7.1228***	0.288	24.744
OilVol_OVX	0.2685***	0.006	43.4335
R^2		0.571	
Adj. R^2		0.571	

Note: Ordinary Least Squares regression with MktVol_VIX as dependent variable and OilVol_OVX as regressor.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.2 Econometric issues

3.2.1 Heteroscedasticity

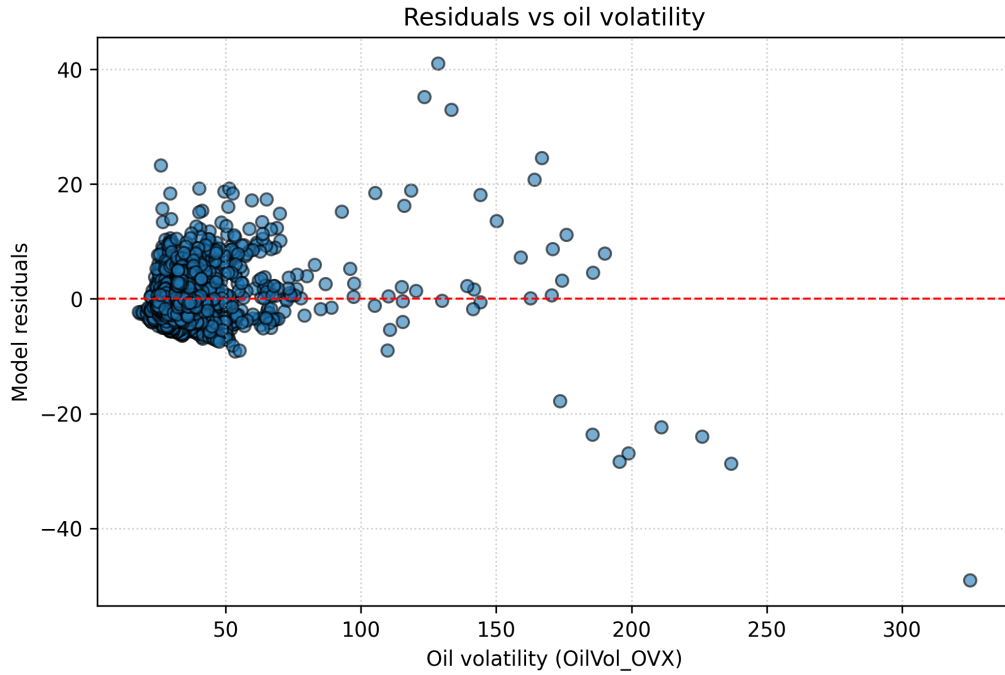
Heteroscedasticity is present, as the variance of the error term may vary across observations. To formally assess this issue, we perform Breusch–Pagan and White tests. Both tests strongly reject the null hypothesis of homoskedasticity, indicating that the variance of the residuals is not constant. A visual inspection of the residual plot (Figure 2) further confirms this pattern, as the dispersion of residuals increases with the level of fitted values. This motivates the use of heteroskedasticity-robust standard errors in subsequent regressions.

Table 2: Tests for Heteroskedasticity

Test	LM Statistic	p-value
White Test	695.10	1.2×10^{-151}
Breusch–Pagan Test	479.79	2.4×10^{-106}

Note: Both tests overwhelmingly reject the null hypothesis of homoskedasticity.

Figure 2: Residual plot showing evidence of heteroscedasticity



3.2.2 Autocorrelation

Autocorrelation may arise due to the daily frequency of the data and the strong persistence commonly observed in volatility dynamics. Serial correlation biases standard errors and leads to inflated statistical significance. The Durbin–Watson statistic is far below 2, indicating severe positive autocorrelation.

Table 3: Durbin–Watson Statistic

Test	Value
Durbin–Watson	0.1953

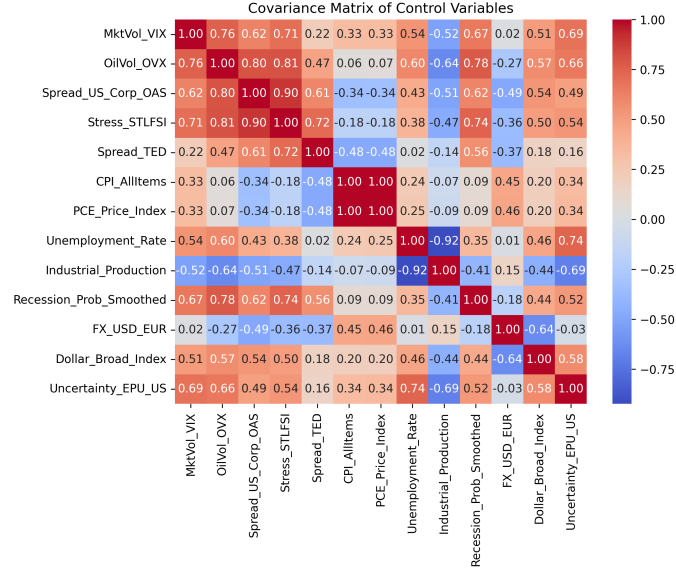
3.2.3 Endogeneity

We also expect our model to suffer from endogeneity, as several macroeconomic variables may be jointly correlated with both the dependent variable and our main regressor. To address this concern, we extend the baseline specification by including a comprehensive set of control variables in an OLS framework.

3.2.4 Multicollinearity

However, introducing multiple controls raises another potential issue: multicollinearity. To assess this, we examine the pairwise correlations among the eleven control variables. We compute a correlation matrix and visualize it through a heatmap, which provides a clear representation of the dependence structure within the vector of controls. The resulting heatmap is displayed in Figure 3.

Figure 3: Correlation heatmap of the control variables



From the heatmap, we observe that `CPI_ALLTERM` and the `PCE_Price_Index` are highly correlated, to the point of being essentially identical. We therefore remove `CPI_ALLTERM` from the set of controls. We also note a strong correlation between `STRESS_STLFSI` and the U.S. corporate spread, which motivates the exclusion of `STRESS_STLFSI` to avoid redundant information.

In addition, we choose to discard variables that exhibit only weak correlation with the dependent variable, in order to preserve degrees of freedom and avoid an excessively large set of regressors with limited explanatory relevance. Following this principle, we remove the `FX_USD_EUR` exchange rate, as it shows minimal correlation with financial market volatility. Finally, given the strong negative correlation between the unemployment rate and industrial production, we retain only one of the two and exclude industrial production.

After applying these selection criteria, we keep a final set of eight control variables for the extended specification.

4 Model specification and optimization

4.1 Model with Additional Control Variables and HAC-Robust Errors

We extend the baseline specification by incorporating a vector of control variables selected on the basis of their acceptable multicollinearity properties. The augmented model is given by:

$$\text{MktVol_VIX}_i = \beta_0 + \beta_1 \text{OilVol_OVX}_i + \gamma' X_i + \varepsilon_i,$$

where X_i denotes the set of eight macro-financial controls included to mitigate potential omitted-variable bias:

$$X_i = \left\{ \begin{array}{l} \text{Stress_STLFSI, Spread_TED, PCE_Price_Index,} \\ \text{Unemployment_Rate, Industrial_Production,} \\ \text{Recession_Prob_Smoothed, Dollar_Broad_Index,} \\ \text{Uncertainty_EPU_US} \end{array} \right\}.$$

After controlling for additional sources of variation and estimating the model with heteroskedasticity- and autocorrelation-consistent (HAC) standard errors, the coefficient of interest decreases slightly—by approximately 0.2%—while its standard error increases. These adjustments align with the expectation that the simple OLS specification may suffer from omitted-variable bias and inefficient inference.

Nonetheless, the coefficient on oil market volatility remains statistically significant at the 5% level. This suggests that the contemporaneous link between oil volatility and financial market volatility is robust to the inclusion of macroeconomic controls and to corrections for potential heteroskedasticity and autocorrelation.

Table 4: OLS with Controls and HAC Robust Standard Errors

	Coefficient	Std. error	t-stat
Constant	-49.4871**	21.3045	-2.320
OilVol_OVX	0.0502**	0.0198	2.530
R^2		0.789	
Adj. R^2		0.787	

Note: OLS regression with MktVol_VIX as dependent variable, including all eight control variables (not reported). HAC (Newey–West) robust standard errors are reported in the second column.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.2 Models with Functional Form Adjustments

To further refine our specification, we explore several alternative functional forms, including interaction effects, quadratic terms, and logarithmic transformations. These extensions aim to capture potential nonlinearities and improve overall model fit.

4.2.1 Interaction Model

Interaction terms allow the effect of one independent variable on the dependent variable to vary with the level of another variable. Based on theoretical considerations and domain-specific knowledge, we select the following interactions involving oil market volatility:

- **OilVol_OVX $\times$ Stress_STLFSI:** When market stress is elevated, it may amplify the transmission of oil market volatility to financial market volatility. However, the data indicate a negative interaction effect, contrary to expectations.

- **OilVol_OVX $\times$ Dollar_Broad_Index:** A stronger U.S. dollar increases the cost of USD-priced commodities for foreign buyers, potentially reducing demand. This tightening effect may amplify the role of oil volatility in explaining financial market volatility. Empirically, this interaction is positive.
- **OilVol_OVX $\times$ Yield_US3M:** When short-term risk-free rates are high, market uncertainty is already elevated, reducing the marginal contribution of oil volatility to overall market volatility. Conversely, when rates are low, oil volatility becomes more influential.

The full interaction model is therefore specified as:

$$\begin{aligned} \text{MktVol_VIX}_i = & \beta_0 + \beta_1 \text{OilVol_OVX}_i + \gamma' X_i \\ & + \delta_1 (\text{OilVol_OVX}_i \times \text{Stress_STLFSI}_i) + \delta_2 (\text{OilVol_OVX}_i \times \text{Dollar_Broad_Index}_i) \\ & + \delta_3 (\text{OilVol_OVX}_i \times \text{Uncertainty_EPU_US}_i) + \varepsilon_i. \end{aligned} \quad (2)$$

None of the interaction terms were significant at 10%. To avoid losing some degrees of freedom, we decided to remove those interaction terms.

4.2.2 Quadratic Specification

We also test a quadratic functional form to evaluate potential nonlinear effects of oil volatility. Although theory suggests that extreme levels of oil volatility could disproportionately affect financial market volatility, the estimated quadratic term is not significant (p-value = 0.165). Given its strong non-significance, we exclude the quadratic term to avoid unnecessary loss of degrees of freedom and preserve statistical power.

The tested model was:

$$\text{MktVol_VIX}_i = \beta_0 + \beta_1 \text{OilVol_OVX}_i + \beta_2 \text{OilVol_OVX}_i^2 + \gamma' X_i + \varepsilon_i. \quad (3)$$

4.2.3 Log-Linear Specification

Because the relationship between oil volatility and market volatility appears multiplicative, we evaluate a log-log specification. To formally compare linear and logarithmic models, we conduct a PE test following the two-step auxiliary regression procedure.

Let:

- $\hat{y}_i$ denote fitted values from the linear model,
- $\widetilde{\log Y}_i$ denote fitted values from the log-linear model.

The auxiliary regressions are:

1. Test 1 (Linear model):

$$y_i = X_i' \beta + \psi_{\text{lin}}(\hat{y}_i - \widetilde{\log Y}_i) + u_i. \quad (4)$$

2. Test 2 (Log-linear model):

$$\log y_i = (\log X_i)' \gamma + \psi_{\log}(\hat{y}_i - \mathbb{E}[\widetilde{\log Y_i}]) + u_i. \quad (5)$$

The PE test yields the following results:

$$\text{PE Linear: } t(\hat{y}) = -6.877, \quad p = 0.0000,$$

$$\text{PE Log-linear: } t(\hat{y}) = 0.215, \quad p = 0.8298.$$

Thus, Test 1 strongly rejects the hypothesis that the linear model fits better, while Test 2 fails to reject the adequacy of the log-linear specification. Together, these results indicate that the log-log model provides a substantially better functional form.

We therefore adopt the following log-log regression, which achieves an adjusted R^2 of 83.3%:

$$\ln(\text{MktVol_VIX}_i) = \beta_0 + \beta_1 \ln(\text{OilVol_OVX}_i) + \gamma' \ln(X_i) + \varepsilon_i. \quad (6)$$

As anticipated from the PE-test, the log-log specification provides a clearer and statistically stronger relationship between oil market volatility and financial market volatility. As reported in Table 4, the elasticity coefficient on $\ln(\text{OilVol_OVX})$ has a value of 0.6944 while being strongly statistically significant. This value seems to indicate that a 1% change in the oil volatility changes the market volatility by 0.6944%. This improvement confirms that a log-log transformation offers a better functional form for capturing the multiplicative nature of the relationship between the two volatilities, reinforcing the conclusion that the logarithmic model is more appropriate for our data.

However, given how difficult this type of study is, that coefficient looked almost too good to be true. Even though we addressed several issues — using HAC standard errors, adding control variables, and testing non-linear functional forms that improved the model — such a strong coefficient made us think that something might still be wrong. The main remaining concern we identified is the simultaneity problem, which we try to address in the next section.

Table 5: Log-Log OLS with Controls and HAC Robust Standard Errors

	Coefficient	Std. error	t-stat
Constant	-29.4303***	5.810	-5.065
$\ln(\text{OilVol_OVX})$	0.6944***	0.102	6.790
R^2		0.833	
Adj. R^2		0.829	

Note: Log-log OLS regression with $\ln(\text{MktVol_VIX})$ as the dependent variable, including all control variables in logarithmic form (not reported). HAC (Newey-West) robust standard errors are reported in the second column.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Instrumental Variables (IV) / 2SLS Specification

5.1 First-stage regression and instrument diagnostics

To address the simultaneity problem, we computed a final regression using an instrumental variable. To meet the requirements of a strong and relevant instrument, we selected a variable that is strongly correlated with oil volatility but not with the broader financial market. We chose the volume of crude oil available on the market.

To verify that our instrument satisfies both exogeneity and strength conditions, we ran a 2SLS regression and checked that the first-stage F-test value was above 10, and that the correlation between the 2SLS residuals and our instrument was weak. These two checks suggest that our instrument is valid. The detailed results can be found in Table 5

Table 6 summarises the instrument diagnostics (first-stage F-statistic, correlation with residuals, etc.).

Table 6: Instrument diagnostics for the IV specification

	Value
Corr(Z , residuals 2SLS)	3.961×10^{-12}
First-stage F-statistic	40.67

Note: This table reports the main diagnostics for the instrument(s) used in the 2SLS estimation.

5.2 2SLS regression results

Thanks to the relevance and exogeneity of the instrument, we proceeded to estimate the 2SLS regression. Interestingly, once we account for simultaneity and potential endogeneity, the coefficient on OilVol_OVX loses its significance entirely. This indicates that the direct effect of oil market volatility on overall market volatility, which appeared strong in the ordinary least squares estimates, is mainly caused by simultaneity bias between oil and financial markets rather than a causal effect. This means that the impact attributed to oil volatility reflects reactions from the market itself and other confounding factors, rather than a true direct effect from oil to broader market volatility. Hence, the instrumental variable approach highlights that failing to control for endogeneity can substantially overstate the role of oil volatility on market volatility.

Table 7: 2SLS Regression Using Crude Oil Volume as an Instrument (Table 5)

	Coefficient	Std. error	t-stat
Constant	-37.458***	6.9483	-5.3908
$\ln(\text{OilVol_OVX})(2SLS)$	-0.9815	0.6418	-1.5293
R^2		0.3670	
Adj. R^2		0.3529	

Note: Two-stage least squares (2SLS) regression using the logarithm of crude oil volume as an instrumental variable for $\ln(\text{OilVol_OVX})$. The dependent variable is $\ln(\text{MktVol_VIX})$. Instrument validity verified by a first-stage F-statistic above 10 and a weak correlation between the 2SLS residuals and the instrument. All control variables are included in logs but not reported. Detailed results are provided in Table 5.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Results and interpretation

We observe from the last log-linear regression that when price volatility increases by 1%, market volatility rises by 0.694%. Moreover, the robust standard error represents 14.68% of our coefficient, and the adjusted R^2 is around 83%, which is satisfactory in terms of estimation. Given the structure of our data, we considered the possibility of controlling for time fixed effects. These fixed effects would allow us to capture, for each period, the specific impact of that period on market volatility, thereby improving accuracy. However, including such a model would reduce our degrees of freedom; although this cost would be limited, it is still real. Nevertheless, our model already includes the variable `recession_Prob_Smoothed` which measures the probability of a recession at a given time t as a ratio. This variable provides a strong indication of the period-specific effect on market volatility, so we avoid incurring the cost in degrees of freedom by implementing time fixed effects for each of the ten years in our dataset.

These findings could suggest that oil market volatility has a direct and measurable impact on financial market volatility, and that this effect is not simply caused by macroeconomic shocks or cyclical patterns captured by recession probabilities. However, our last model indicates that the exogenous impact of the oil volatility on the market volatility is not significant. Moreover, the estimate coefficient $\ln(\text{OilVol_OVX})$ becomes negative with the 2SLS which demonstrate the strong endogenous factors that influenced our log-log specification. That final estimation makes us reject the null hypothesis and leads us to conclude that we cannot infer any impact of oil volatility on market volatility with the tools we used in this study.

Ultimately, the multiple control variables that we used are still somehow multicollinear, even though we filtered them by testing their multicollinearity. It has to be said that the remaining multicollinearity between our controls may underestimate the standard error of our estimator, which could lead us to reject the null hypothesis when we actually should not.

7 Conclusion

This work was part of a reflection on the relationship between volatility in the oil market and volatility in the US financial market, from the perspective of simultaneous risk transmission. The

study was based on the idea that oil, as a strategic resource and pillar of the global economy and geostrategic relations, has a decisive influence on the expectations and behavior of financial actors. Understanding the interactions between these two markets is a major challenge for forecasting and managing systemic risk. Using an econometric approach based on measuring instantaneous co-movements between oil price volatility and VIX volatility, we sought to assess the oil market's ability to predict, almost instantaneously, the volatility of the US financial market.

Despite our final coefficient being not statistically significant, it remains informative. The estimation seems to be exclusively exogenous as our instrument meet all the requirements that are necessary for a strong instrument. In our context, the weak significance of our coefficient reflect the structural difficulty to isolate an exogenous variation that is strong enough to be considered significant. Our analysis is limited by our academic background and our computational capabilities. We however suggest that the study could be continued by using advanced techniques allowing to correct multiple bias sources. For example, we could control errors dependence by using variance-covariances matrices that are based on GARCH models or standard clusterised errors that are strongly recommended for financial data due to their strong volatility.